Divergences: robust objectives and the KL limit

The generalized-Bayes objective eq. 2 has exactly two tunable ingredients: the divergence D that regularizes the update toward the prior or cavity, and the loss L that measures data fidelity.

Divergences: robust... (part 2)

This section develops both — the divergence family first, the robust loss family in sec. 8.7 — and shows that each carries a limit in which it collapses to its standard counterpart,

KL for the divergence and NLL for the loss.

Divergences: robust... (part 3)

Those client-side limits establish recovery to standard Bayes. The distinct categorical server bridge in sec. 8.8 then identifies a qualified log-linear message-combination specialization; it does not recover the complete source belief-sharing protocol.

Divergences: robust... (part 4)

The regularizing divergence D decides how far a client's updated belief may move from its cavity, so choosing D is a modeling decision rather than a numerical detail. The family lives in `divergences.py`. We implement the forward KL (the standard-Bayes case), the reverse KL (FedGVI's `RKL` client divergence),

Divergences: robust... (part 5)

the standard α-Rényi diagnostic, FedGVI's Alpha-Rényi normalization (AR), and total variation (a bounded distance in $[0, 1]$).

The single most important recovery property is that the robust members recover the KL divergence in a limit:

Divergences: robust... (part 6)

$$D_\alpha(q \,\|\, p) \xrightarrow[\alpha \to 1]{} \mathrm{KL}(q \,\|\, p). \tag{6}$$

Divergences: robust... (part 7)

Lemma (KL is the $\alpha \to 1$ limit of the Rényi family)

For finite-support categorical pmfs q, p, $D_\alpha(q \,\|\, p) = (\alpha - 1)^{-1} \log \sum_k q_k^\alpha p_k^{1-\alpha}$ tends to $\mathrm{KL}(q \,\|\, p)$ as $\alpha \to 1$, the limit (6). Near $\alpha = 1$, `divergences.py` returns the KL closed form, making equality exact there rather than asymptotic.

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Divergences: robust... (part 8)

KL is the divergence that makes generalized Bayes collapse to standard Bayes. When local posteriors are then combined by the separately specified categorical message-combination specialization in sec. 8.8, the project recovers its log-linear-pool corner;

Divergences: robust... (part 9)

neither step reconstructs the complete belief-sharing protocol of Friston et al. (Friston et al. 2024).

Divergences: robust... (part 10)

Everything robust is a controlled departure from that fixed point; Lemma 3 is the formal hinge, and the largest observed Rényi-versus-KL discrepancy in the recovery band is 0 (reported in sec. 10.1).

Divergences: robust... (part 11)

The standard Rényi diagnostic is `renyi_divergence`; FedGVI's `AR` regularizer is `alpha_renyi_divergence`, equal to the standard form divided by α. For the finite categorical support,

Divergences: robust... (part 12)

`generalized_posterior` solves the named Alpha-Rényi objective through its scalar normalization condition rather than using a generic power-softmax shortcut.

Divergences: robust... (part 13)

This distinction keeps the reported limit and the implemented objective aligned. The `AR` regularizer is not merely a diagnostic: it is exercised as a client divergence in the categorical FedGVI baseline of sec. 10.6,

Divergences: robust... (part 14)

where it pairs with the rcce loss of sec. 8.7 to constitute the genuine per-client robustness axis.

Robust losses: bounded influence at the Bayes corner

The data-fidelity term of eq. 2 lives in `losses.py`. Standard Bayes uses the negative log-likelihood, $\mathrm{NLL}(p, o) = -\log p(o)$, which is *unbounded*:

Robust losses (part 2)

a single contaminated observation with $p(o) \to 0$ dominates the posterior.

Robust losses (part 3)

This is precisely the fragility the robust-Bayes literature was built to remove (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015; Zhang and Sabuncu 2018), extended into robust-divergence variational inference (Futami et al. 2018), and the property FedGVI imports into federated inference (Mildner et al. 2025).

Robust losses (part 4)

The robust-statistics vocabulary here is the usual influence-function one (Huber and Ronchetti 2009): bounded losses reduce the leverage of extreme observations, while NLL does not.

Robust losses (part 5)

We implement two categorical robust losses, each of which recovers NLL in a limit.

The density-power (β) loss (Basu et al. 1998; Fujisawa and Eguchi 2008; Ghosh and Basu 2015; Futami et al. 2018) is recentered so that the scalar limit is exact:

Robust losses (part 6)

$$L_\beta(p, o) = -\frac{p(o)^\beta - 1}{\beta} + \frac{\sum_k p_k^{\beta+1} - 1}{\beta + 1}, \quad (7)$$

$$L_\beta \xrightarrow[\beta \to 0]{} \text{NLL}.$$

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Robust losses (part 7)

The robust categorical cross-entropy (generalized cross-entropy) (Zhang and Sabuncu 2018) is

Robust losses (part 8)

$$L_{q_{\text{loss}}}(p, o) = \frac{1 - p(o)^{q_{\text{loss}}}}{q_{\text{loss}}}, \qquad L_{q_{\text{loss}}} \xrightarrow[q_{\text{loss}} \to 0]{} \text{NLL}. \tag{8}$$

Robust losses (part 9)

which by l'Hôpital recovers NLL as $q_{\text{loss}} \to 0$ and at $q_{\text{loss}} = 1$ is the bounded mean-absolute-error loss $1 - p(o)$, finite exactly where NLL diverges.

Robust losses (part 10)

Proposition (β-loss and rcce recover NLL)

The losses in (7) and (8) recover NLL as their respective loss parameters tend to zero. Both limits are exact in the implementation. At their bounded endpoints, both losses remain finite where NLL diverges.

Robust losses (part 11)

The largest observed discrepancies are 0 and 0, respectively (sec. 10.1). sec. 10.5 validates the bounded-loss robustness.

Robust losses (part 12)

Taking the loss-parameter limits ($\beta \to 0$ or $q_{\text{loss}} \to 0$) reproduces standard Bayes. Combining those local posteriors through the qualified categorical specialization in sec. 8.8 is a separate server step, not a recovery claim for the complete belief-sharing protocol of Friston et al. (2024).

Robust losses (part 13)

This is the **per-agent rigorous robustness axis**: it is derived from eq. 2 and provably limits to Bayes through Proposition 4 and Lemma 3,

Robust losses (part 14)

and it is the axis that carries FedGVI's bounded-influence guarantee under the cited matching assumptions.

Robust losses (part 15)

The complementary server-side divergence-reweighting heuristic of sec. 8.8 is a distinct device and is never granted this guarantee (sec. 6.3).

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